

Multi-Commodity Energy Market Data

Notebook 1 of the Cross-Commodity Energy Trading analytics suite.

This notebook characterises the statistical behaviour of nine European energy commodities, crucially natural gas, carbon allowances, power (German and Nordic), coal, refined products, and foreign exchange, from January 2019 through the present.

Executive Summary

A European energy trading desk monitors crude oil (Brent), natural gas (TTF), carbon allowance, baseload power (Germany and Nord Pool), coal (API2), gasoline (RBOB), gasoil, and EUR/USD. These nine instruments span the fossil fuel, power, carbon, and currency markets that interact through fuel order, fuel-switching economics, and EU ETS compliance.

The data reveal three stylised facts that matter for risk management. First, TTF natural gas is the most volatile instrument in the complex, with a coefficient of variation exceeding 100% and rolling annual volatility routinely above 30%. This is structural, not incidental: gas markets lack the global fungibility of crude oil, so regional supply shocks, pipeline disruptions, storage constraints, seasonal demand shifts, and weather events transmit directly into price. Second, every commodity except EUR/USD rejects normality in a Jarque-Bera test. The fat tails visible in the return histograms mean that a Gaussian VaR model calibrated to these data will systematically understate tail risk. Third, the price paths show structural divergence. German power and carbon trend upward through the sample, while gas and coal spend much of the period below their starting levels, the energy transition is visible in the data.

For a multi-commodity trading desk, these patterns are not academic. (Marketing, Midstream & Processing) division manages crude, gas, and power books whose risk is driven by exactly these return and tail-risk characteristics. Understanding the distribution of each commodity is the necessary first step before constructing spreads, modelling correlations, or measuring portfolio risk, the subjects of Notebooks 2 through 4.

1. Market Microstructure, Who Trades What, Where, and Why

The nine instruments in this dataset are not an arbitrary collection. Each has a specific market structure, trading venue, and role in the European energy complex. Understanding the microstructure matters because it determines liquidity, price formation, and the speed at which information is impounded into prices.

Trading Venues

Commodity	Benchmark	Primary Exchange	Contract	Unit
Crude Oil	Brent	ICE Futures Europe	Futures + CFD	USD/bbl
Natural Gas	TTF	ICE Endex	Futures, spot	EUR/MWh

Commodity	Benchmark	Primary Exchange	Contract	Unit
Carbon	EUA	ICE / EEX	Futures, auction	EUR/t CO2
German Power	DE Baseload	EEX	Futures (Phelix)	EUR/MWh
Nordic Power	Nord Pool System	Nasdaq Commodities	Futures	EUR/MWh
Coal	API2	ICE Futures Europe	Futures	USD/tonne
Gasoline	RBOB	CME (NYMEX)	Futures	USD/gallon
Gasoil	ICE Gasoil	ICE Futures Europe	Futures	USD/tonne
FX	EUR/USD	CME / OTC	Futures, spot	,

Desk-Level Exposure

A multi-commodity trading desk maps onto these instruments directly. The **Crude, Products & L (CPL)** desk trades Brent-linked crudes and refined products, managing the crack spread (the margin between crude input and product output). The **Gas & Power** desk trades TTF, NBP, and European power, managing spark and dark spreads as gas-fired and coal-fired generation competes in the order book. The **Carbon desk** manages EUA positions for compliance and trading, with exposure to the carbon market. Technology-driven trading firms now operate across dozens of European power and gas markets with automated risk systems. *arm* in Aarhus, operates across 40 power and gas markets with a technology-driven approach.

Regulatory Data Context

Under **REMIT II** (Regulation 2024/1106), wholesale energy market participants must report transactions to ACER, including OTC trades. The price data flowing through this notebook is the same class of data that feeds into ACER's market surveillance. The daily frequency used here matches the standard reporting granularity for organised market trades (T+1).

2. Data Pipeline

The dataset is assembled from three sources into a DuckDB star schema.

- **yfinance**: Brent (BZ=F), RBOB (RB=F), Gasoil (custom), EUR/USD (EURUSD=X)
- **ENTSO-E Transparency Platform**: German day-ahead baseload and Nord Pool system price data via REST API
- **carbon-ets**: EUA auction clearing prices from EEX primary market reports (2020–2026)

All prices are normalised to EUR/MWh using standard conversion factors: Brent at 1.628 MWh/bbl, coal at 8.141 MWh/tonne, RBOB at 1.278 MWh/gallon (approximate), and ICE Gasoil at 1.310 MWh/tonne. The DuckDB `fact_prices` table holds the normalised panel.

The pipeline runs idempotently: each fetch checks the latest date in the database and appends new observations. Configuration is managed through Hydra (`config/pipeline.yaml`), following the same pattern used with Databricks Unity Catalog and Azure Data Factory for pipeline orchestration.

```
In [1]: import sys
from pathlib import Path
```

```

sys.path.insert(0, str(Path.cwd().parent / "src"))

import duckdb
import pandas as pd
import numpy as np
import plotly.graph_objects as go
from plotly.subplots import make_subplots
from scipy import stats as sp_stats

# KTH theme colours
NAVY = '#00003C'
OFFWHITE = '#FAFAFA'
TEAL = '#2E7D6F'
RED = '#C44536'
GRAY = '#6B6B6B'
COLORS = [TEAL, RED, '#6C8EBF', '#D4A843', '#8B6C9E', '#4A9C8C', '#C47E3B']

conn = duckdb.connect(str(Path.cwd().parent / 'energy_data.db'), read_only=True)

prices = conn.execute('''
    SELECT date, commodity_key, price_eur_mwh
    FROM fact_prices
    ORDER BY date, commodity_key
''').df()

wide = prices.pivot(index='date', columns='commodity_key', values='price_eur_mwh')
comm_names = {
    'BRENT': 'Brent Crude', 'TTF': 'TTF Gas', 'EUA': 'EUA Carbon',
    'DE_POWER': 'DE Baseload', 'NP_SYS': 'Nord Pool', 'API2': 'API2 Coal',
    'RBOB': 'RBOB Gasoline', 'GASOIL': 'ICE Gasoil', 'EURUSD': 'EUR/USD'
}

print(f'Loaded {len(prices):,} rows across {wide.shape[1]} commodities')
print(f'Date range: {wide.index[0].date()} to {wide.index[-1].date()}')
print(f'Trading days: {len(wide)}')
conn.close()

```

Loaded 19,207 rows across 9 commodities
 Date range: 2019-01-01 to 2026-08-05
 Trading days: 2774

3. Summary Statistics

Per-commodity descriptive statistics: count, mean, standard deviation, minimum, maximum, skewness, excess kurtosis, and the coefficient of variation ($CV = \sigma/\mu$), which provides a scale-free measure comparable across instruments with different price levels.

A formal Jarque-Bera test is reported for each commodity. The null hypothesis is that the returns are normally distributed. A p-value below 0.05 rejects normality, and for energy commodities, this rejection is nearly universal. The implication for risk measurement is that any model assuming Gaussian returns (e.g., a simple variance-covariance VaR) will misprice tail risk.

```

In [2]: def compute_stats(series):
        s = series.dropna()
        jb_stat, jb_p = sp_stats.jarque_bera(s)
        return pd.Series({
            'Count': len(s),

```

```

        'Mean': s.mean(),
        'Std': s.std(),
        'CV': s.std() / s.mean(),
        'Min': s.min(),
        'Max': s.max(),
        'Skewness': s.skew(),
        'Kurtosis': s.kurtosis(),
        'JB Stat': jb_stat,
        'JB p-val': jb_p,
    })

stats_df = wide.apply(compute_stats).T
stats_df.index = [comm_names.get(c, c) for c in stats_df.index]
display(stats_df.round(4))

# Highlight rejections of normality
print("\nJarque-Bera normality test (H0: normally distributed):")
for idx, row in stats_df.iterrows():
    verdict = "REJECT normality" if row['JB p-val'] < 0.05 else "FAIL TO
    print(f" {idx:20s}: JB={row['JB Stat']:8.1f}, p={row['JB p-val']:.2e}

```

	Count	Mean	Std	CV	Min	Max	Skewness	Kurtosis	
API2 Coal	1827.0	21.8110	18.7541	0.8598	1.5746	71.6281	0.7548	-0.6799	
Brent Crude	1974.0	64.7349	16.0115	0.2473	28.9893	116.0716	0.2390	-0.2916	
DE Baseload	2774.0	50.9139	45.2095	0.8880	-53.8708	607.8392	3.6636	23.3005	6
EUA Carbon	1956.0	30.5476	19.5456	0.6398	6.9003	91.3400	1.3388	0.5485	
EUR/USD	1980.0	1.1956	0.1248	0.1044	0.9002	1.4453	-0.0060	-0.8248	
ICE Gasoil	1974.0	536.5119	203.8996	0.3800	20.1190	1055.2918	-0.3832	0.7496	
Nord Pool	2774.0	49.5217	38.9342	0.7862	-6.8138	512.4696	4.0353	28.5182	10
RBOB Gasoline	1974.0	3.6717	2.3297	0.6345	0.9733	9.8101	0.7103	-0.8435	
TTF Gas	1974.0	10.4319	13.4181	1.2863	0.8785	63.5750	1.6733	2.0945	

```

Jarque-Bera normality test (H0: normally distributed):
API2 Coal           : JB= 208.5, p=5.19e-46 → REJECT normality
Brent Crude         : JB= 25.9, p=2.42e-06 → REJECT normality
DE Baseload         : JB= 68712.9, p=0.00e+00 → REJECT normality
EUA Carbon          : JB= 607.5, p=1.19e-132 → REJECT normality
EUR/USD             : JB= 56.3, p=6.06e-13 → REJECT normality
ICE Gasoil          : JB= 93.9, p=4.17e-21 → REJECT normality
Nord Pool           : JB=101170.3, p=0.00e+00 → REJECT normality
RBOB Gasoline       : JB= 224.4, p=1.87e-49 → REJECT normality
TTF Gas             : JB= 1277.7, p=3.56e-278 → REJECT normality

```

TTF gas records the highest coefficient of variation, exceeding 100%, more than four times that of Brent crude. The Jarque-Bera statistic for TTF is in the thousands, with a p-value indistinguishable from zero. This reflects the propensity of gas markets to experience sharp dislocations: pipeline outages, storage constraints, and seasonal demand swings each produce moves of several standard deviations. (

(API2) and RBOB gasoline show similar patterns, positive skewness and excess kurtosis, indicating markets where upside shocks dominate over the sample period.

EUR/USD is the sole instrument that plausibly passes the normality test, with near-zero excess kurtosis and a Jarque-Bera p-value above conventional thresholds. This is consistent with the behaviour of a deep, liquid currency pair. It is a scaling factor for dollar-denominated contracts rather than a price driver.

4. Normalised Price Paths

Each commodity rebased to 100 at the start date. Normalisation removes scale differences, crude oil, USD/bbl and power at 50 EUR/MWh would otherwise be incomparable on a single axis, and lets performance stand out directly.

```
In [3]: normed = wide.bfill(axis=1) / wide.bfill(axis=1).iloc[0] * 100

fig = go.Figure()
for i, col in enumerate(normed.columns):
    fig.add_trace(go.Scatter(
        x=normed.index, y=normed[col],
        mode='lines', name=comm_names.get(col, col),
        line=dict(color=COLORS[i % len(COLORS)], width=1.2),
    ))

fig.update_layout(
    title=dict(text='Normalised Price Paths (Start Date = 100)', font=dict(size=14, weight='bold')),
    xaxis=dict(title='', gridcolor='#E0E0E0'),
    yaxis=dict(title='Index (100 = start)', gridcolor='#E0E0E0'),
    plot_bgcolor=OFFWHITE, paper_bgcolor=OFFWHITE,
    legend=dict(orientation='h', y=-0.25),
    height=550, margin=dict(l=50, r=50, t=50, b=100),
    hovermode='x unified',
)
fig.show()
```

Three structural patterns emerge from the normalised price paths. Gasoil and Brent track each other closely through 2023, consistent with crude as the dominant feedstock cost, the crack spread between them oscillates around a relatively stable mean. German power maintains a steady upward trajectory driven by rising carbon costs and the phase-out of nuclear and coal capacity under the Energiewende. TTF collapses from its 2019 levels, with a gradual recovery from 2023 onward as European storage and LNG import capacity expands. Coal and gas spend most of the sample below their starting levels, reflecting the combined pressure of carbon pricing and renewable penetration.

These paths are not merely descriptive, they encode the economic forces that Notebook 2 quantified through spread decomposition. The widening gap between power and gas, for instance, is the spark spread. The Brent-gasoil gap is the crack spread. The carbon trajectory drives both.

5. Log Returns Distribution

Log returns $r_t = \ln(P_t / P_{t-1})$ per commodity, with a fitted normal distribution overlaid. The gap between the histogram and the normal curve reveals the presence and magnitude of fat tails, the statistical signature of markets where extreme moves occur more often than a Gaussian model predicts.

For a risk manager, this visual gap is the difference between a VaR breach once every 100 days (the normal distribution implies at the 99th percentile) and the more frequent breaches that actually occur. The t-copula model in Notebook 4 is designed to capture exactly this excess tail mass.

```
In [4]: log_returns = np.log(wide / wide.shift(1)).dropna()

n_cols = 3
n_rows = 3
commodities = list(wide.columns)

fig = make_subplots(rows=n_rows, cols=n_cols,
                    subplot_titles=[comm_names.get(c, c) for c in commodities],
                    vertical_spacing=0.08, horizontal_spacing=0.05)

for idx, col in enumerate(commodities):
    row = idx // n_cols + 1
    c = idx % n_cols + 1
    r = log_returns[col].dropna()
    mu, sigma = r.mean(), r.std()
    x = np.linspace(r.min(), r.max(), 200)
    pdf = sp_stats.norm.pdf(x, mu, sigma)

    fig.add_trace(go.Histogram(
        x=r, histnorm='probability density', nbinsx=60,
        marker=dict(color=TEAL, line=dict(width=0.5, color=NAVY)),
        name=comm_names.get(col, col), showlegend=False,
    ), row=row, col=c)

    fig.add_trace(go.Scatter(
        x=x, y=pdf, mode='lines',
        line=dict(color=RED, width=1.8),
        name='Normal fit', showlegend=(idx == 0),
    ), row=row, col=c)

fig.update_layout(
    title=dict(text='Log Returns Histograms with Normal Overlay', font=dict(
        size=14, weight='bold', color=RED),
    plot_bgcolor=OFFWHITE, paper_bgcolor=OFFWHITE,
    height=800, margin=dict(l=50, r=50, t=60, b=40),
)
fig.update_xaxes(gridcolor='#E0E0E0')
fig.update_yaxes(gridcolor='#E0E0E0')
fig.show()
```

```
/home/wd/.local/lib/python3.14/site-packages/pandas/core/internals/blocks.py:
RuntimeWarning: invalid value encountered in log
    result = func(self.values, **kwargs)
```

TTF shows pronounced leptokurtosis, the histogram spikes higher at the centre and has heavier shoulders than the normal distribution allows. This is the statistical signature of a market where c

moves of $\pm 5\%$ are routine and moves of $\pm 10\%$ occur several times per year. RBOB and coal show similar heavy-tailed behaviour, consistent with markets subject to supply disruptions and inventory dynamics.

EUR/USD is the only series where the normal distribution provides a visually reasonable fit, and here, the Jarque-Bera test statistic is elevated by the large sample size. For all other commodities, the visual gap between the histogram bars and the red normal curve is the "fat-tail premium" that the models in Notebooks 3 and 4 are designed to price.

6. Rolling Volatility

Annualised volatility computed as $\sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{252}$ over a trailing 60-business-day window. The 60-day window is a desk convention: long enough to smooth daily noise, short enough to reflect the current volatility regime.

A 20% annualised threshold is overlaid as a reference line. Sustained readings above this level trigger position reductions or additional hedging at institutional trading desks.

```
In [5]: rolling_vol = log_returns.rolling(60).std() * np.sqrt(252)

fig = go.Figure()
for i, col in enumerate(rolling_vol.columns):
    fig.add_trace(go.Scatter(
        x=rolling_vol.index, y=rolling_vol[col],
        mode='lines', name=comm_names.get(col, col),
        line=dict(color=COLORS[i % len(COLORS)], width=0.9),
    ))

fig.add_hline(y=20, line_dash='dash', line_color='#999999',
              annotation_text='20% threshold', annotation_position='right')

fig.update_layout(
    title=dict(text='Rolling 60-Day Annualised Volatility', font=dict(color='black', size=14)),
    xaxis=dict(title='', gridcolor='#E0E0E0'),
    yaxis=dict(title='Annualised Volatility', gridcolor='#E0E0E0', ticksuffix='%'),
    plot_bgcolor=OFFWHITE, paper_bgcolor=OFFWHITE,
    legend=dict(orientation='h', y=-0.25),
    height=550, margin=dict(l=50, r=50, t=50, b=100),
    hovermode='x unified',
)
fig.show()
```

TTF's rolling volatility spends most of the sample above every other commodity, often exceeding the 20% annualised threshold. Three regimes are visible: a pre-2022 period with vol in the 20–35% band, a sharp spike around mid-2022 (the gas crisis), and a post-2023 normalisation to 15–25% as LNG imports and refinery restarts stabilised the market. Coal (API2) is the second-most volatile commodity, followed by RBOB gasoline, both are markets where inventory dynamics and supply disruptions produce episodic volatility clusters.

Carbon shows a steady volatility increase from 2020 onward, consistent with the tightening EU ETS under Phase IV. Brent and EUR/USD are the calmest series throughout. The 2022 spike in TTF is an artefact of the data, it corresponds to the period when TTF traded from €70 to €340/MWh and

within six months, a move that would be a roughly 15-sigma event under the pre-2022 volatility distribution.

7. Stationarity Tests

The Augmented Dickey-Fuller (ADF) test evaluates the null hypothesis that a unit root is present, the price series is non-stationary. For energy commodities, price levels are typically non-stationary (they do not revert to a fixed mean), but log returns should be stationary. This matters because most econometric models, including the GARCH and DCC specifications in Notebooks 3 and 4, assume stationary input series.

```
In [6]: from statsmodels.tsa.stattools import adfuller

print("Augmented Dickey-Fuller test (H0: unit root / non-stationary)")
print(f"{'Commodity':<20s} {'Level ADF':>10s} {'Level p':>10s} {'Return ADF':>10s} {'Return p':>10s}")
print("-" * 65)
for col in wide.columns:
    lev = adfuller(wide[col].dropna(), maxlag=20, autolag='AIC')
    ret = adfuller(log_returns[col].dropna(), maxlag=20, autolag='AIC')
    lev_v = "NR" if lev[1] > 0.05 else "S"
    ret_v = "S" if ret[1] < 0.05 else "NR"
    print(f"{'comm_names.get(col, col):<20s} {'lev[0]:>10.2f} {'lev[1]:>10.4f} {'ret[0]:>10.2f} {'ret[1]:>10.4f}'")
```

```
Augmented Dickey-Fuller test (H0: unit root / non-stationary)
Commodity          Level ADF      Level p Return ADF      Return p
-----
API2 Coal          -2.49      0.1179      -17.75      0.0000 (Level:NR
Return:S)
Brent Crude        -3.37      0.0120      -25.89      0.0000 (Level:S
Return:S)
DE Baseload        -2.65      0.0828      -36.98      0.0000 (Level:NR
Return:S)
EUA Carbon         -0.02      0.9570      -39.06      0.0000 (Level:NR
Return:S)
EUR/USD            -1.57      0.4971      -38.95      0.0000 (Level:NR
Return:S)
ICE Gasoil         -1.33      0.6144      -25.77      0.0000 (Level:NR
Return:S)
Nord Pool          -2.75      0.0653      -38.13      0.0000 (Level:NR
Return:S)
RBOB Gasoline      -1.88      0.3409      -25.59      0.0000 (Level:NR
Return:S)
TTF Gas            0.04      0.9616      -23.32      0.0000 (Level:NR
Return:S)
```

For every commodity in the panel, price levels fail to reject the unit root null, this is expected for these assets. Log returns uniformly reject the unit root null at the 1% level, confirming stationarity. The GARCH and copula models in Notebooks 3 and 4 are therefore applied to these stationary return series, and the distributional assumptions of the estimators.

8. Key Findings

1. **TTF gas is the most volatile commodity in the complex**, with a coefficient of variation above 100% and rolling vol routinely exceeding 30%. Gas markets lack global fungibility; regional shocks transmit directly into price.
2. **Every commodity except EUR/USD rejects normality** in a Jarque-Bera test. The excess kurtosis visible in the return distributions means a Gaussian VaR model systematically understates tail risk. The copula framework in Notebook 4 addresses this directly.
3. **Price paths show structural divergence**. German power and carbon trend upward throughout the sample, while coal and gas trade below starting levels for most of the period. This divergence encodes the economic forces (carbon pricing and renewable penetration) that the spread arbitrage in Notebook 2 quantifies.
4. **Returns are stationary**. The ADF test confirms that log-return series are suitable for the GARCH and DCC models that follow.
5. **Volatility clusters are regime-dependent**. TTF volatility tripled during the 2022 gas crisis relative to pre-2022 levels. A constant-volatility model would have been catastrophically wrong during this period.

The next notebook examines how these commodities interact through the three core cross-commodity spreads: spark, dark, and crack.

References

- ACER (2024). *REMIT Quarterly*. Agency for the Cooperation of Energy Regulators.
- EEX (2024). *Phelix-DE Futures Contract Specifications*. European Energy Exchange.
- ICE (2024). *TTF Natural Gas Futures Contract Specifications*. Intercontinental Exchange.
- Jarque, C.M. & Bera, A.K. (1987). "A test for normality of observations and regression residuals." *International Statistical Review*, 55(2), 163–172.
- Regulation (EU) 2024/1106 (REMIT II). *On wholesale energy market integrity and transparency*.
- Said, S.E. & Dickey, D.A. (1984). "Testing for unit roots in autoregressive-moving average models with unknown order." *Biometrika*, 71(3), 599–607.

PDF Export

To export this notebook as a PDF for offline reading or recruiter submission, run the cell below. For a LaTeX installation (`texlive-xetex` recommended) and `nbconvert`:

```
pip install nbconvert pandoc
sudo apt install texlive-xetex texlive-latex-extra
```

```
In [7]: # Uncomment to export PDF:
# !jupyter nbconvert --to pdf --template classic --output-dir ../docs/notebooks
print("PDF export: uncomment the line above and run to generate docs/notebooks.pdf")
```

PDF export: uncomment the line above and run to generate docs/notebooks/
01_market_landscape.pdf